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Antenna dan Propagasi

UTS

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① $f(\theta,\phi) = 2\cos^4\theta$ dimana $0 \le \theta \le 90°$; $0 \le \phi \le 360°$

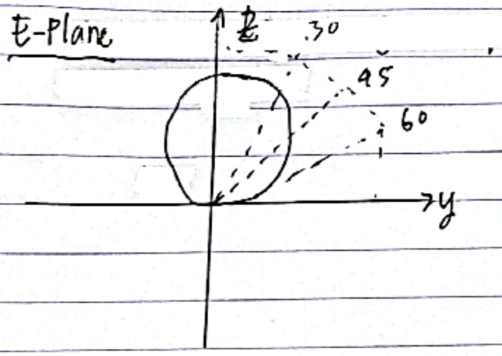
a.) sketsa E-plane dan H-plane

$f(\theta,\phi) = 2\cos^4\theta$

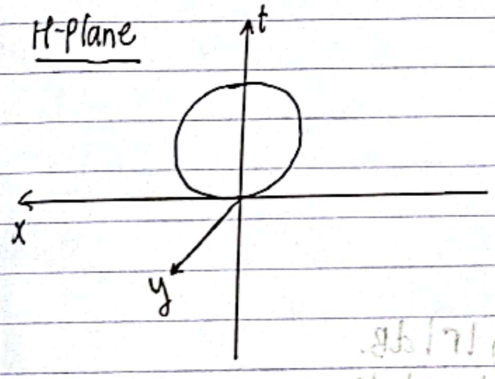
$2\cos^4\theta = \frac{\sqrt{2}}{2}$

$\theta = \cos^{-1}\left(\sqrt[4]{\frac{\frac{\sqrt{2}}{2}}{2}}\right) = 39,55°$

E-Plane



H-Plane



b.) Hitung HPBW y E-plane
H-plane dalam radian

HP-1 = $2\theta = 2(39.55°) = 79,1°$

HP-2 = $2\theta = 2(39.55) = 79,1°$

c.) $D = \frac{41253}{79,1° \cdot 79,1°}$

$= 6,62°$

d.) Unidireksional

② DSLL = 20dB. Level side lobe = 15dB.

a.) Selisih antara main lobe dan side lobe dalam satuan linier dan desibel

$SLL = \frac{\text{lvl side lobe}}{\text{main lobe}}$

$20 = \frac{15}{\text{main lobe}}$

Main lobe = $\frac{15}{20} = 0,75$ dB.

Selisih = $15 - 0,75 = 14,25$ dB.

$10^{\frac{14,25}{10}} = 26.6$ (selisih dlm linier)

FBR = 13dB. lvl. back lobe 3mW, lvl main lobe.

b.) Back lobe = 3 mW

$= 10 \cdot \log 3\text{mW}$

$= 4,77$ dBm

$FBR = \frac{\text{main lobe}}{\text{back lobe}}$

$13 = \frac{\text{main lobe}}{4,77}$

main lobe $= 13 \cdot (4,77)$

$= 62,01$ dBm.

④ Antena susun kolinier 7-elemen dipol ideal pada frek 2400 – 2500 MHz

a.) Jarak antar elemen

$$\lambda = \frac{c}{F} = \frac{3.10^8}{2,4 \times 10^9} = 0,125\ m = 125\ mm.$$

Jarak antar elemen

$$\text{to } \frac{\lambda}{2} = \frac{0,125\ m}{2} = 0,625\ m.$$

b. Gain antena susun

$$W = \frac{c}{2f}\sqrt{\frac{2}{Gr+1}} = \frac{3.10^8}{2.2,4 \times 10^6}\sqrt{\frac{2}{4,3+1}} = 0,030\ m = 38\ mm.$$

Gainnya.

$$G_1 = \frac{1}{g_0}\left(\frac{W}{\lambda_0}\right)^2 = \frac{1}{g_0}\left(\frac{38}{124}\right)^2 = 1,02 \times 10^{-3}$$ //

c.) VSWR 1,5 : 1. Presentase daya yang dikembalikan o/ antena kearah perangkat radio

$$VSWR = \frac{1 + |r|}{1 - |r|}$$

$$1,5 = \frac{1 + r}{1 - r}$$

$$1,5(1-r) = 1 + r$$

$$1,5 - 1,5r = 1 + r$$

$$2,5r = 0,5$$

$$r = \frac{0,5}{2,5}$$

$$r = \frac{1}{2} \cdot \frac{2}{5}$$

$$r = \frac{1}{5} = 0,2\ dB.$$

$$RL = -20 \log |r|\ dB.$$

$$= -20 \log |0,2|\ dB$$

$$RL = 13,97\ dB$$

$$10.\ \frac{13,97}{10} = 24,34. \Rightarrow \frac{1}{24,34}\ (\text{dr pantul})$$

$$\% \Rightarrow \frac{1}{24,34} \times 100\%$$

$$= 4,1\ \%$$

d.) Susunan antena (sketsa grafik. e.)



(3.) P_T watt ; Gain$_T$ = 70 ; efisiensi = 85% . F = 1,2 GHz . r = 7 km
↳ 10.log70 = 18,45 dB → Redaman = 15dB/100 m (15meter)
↳ 2,25 dB

$P_R = 10^{-6}$ mW ; G_R = 50 ; efisiensi = 90% ; panjang = 1,5m .
↳ 10. log 10^{-6} = -60 dBm ↳ G_R = 10 log 50 = 17dB
Redaman = (15dB/100m) → 1,5 dB

P_R = -60 - 2,25 = -62,25 dBm

-62,25 = P_T + 18,45 + 17 - 20 log 7 - 20 log 1200 - 32,45
-62,25 = P_T + 35,45 - 16,9 - 61,6 - 32,45
-62,25 = P_T - 75,5
P_T = 75,5 - 62,25
P_T = 13,25
13,25 - 1,5 = 11,75 dBm
$10^{\frac{11,75}{10}}$ = 14,96 mW.

Bintang Obor